

Chapter 26: Currents

Free electrons: in metals

Ions: in electrolysis (заряженные атомы/молекулы)

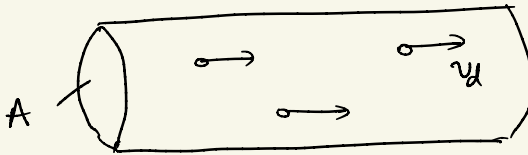
Wood, glass

vs

copper

free electron

holes (отсутствие электронов)



n : number of charge carriers

$$\boxed{I = nq\mu A} = nqv_d A$$

$$e = -1.602 \times 10^{-19} \text{ C}$$

$$J = \frac{I}{\text{Area}}$$

плотность тока

$$\sigma = \frac{Q}{\text{Area}}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0} = \int \frac{\rho d\tau}{\epsilon_0}$$

$$\text{in the future: } \oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$

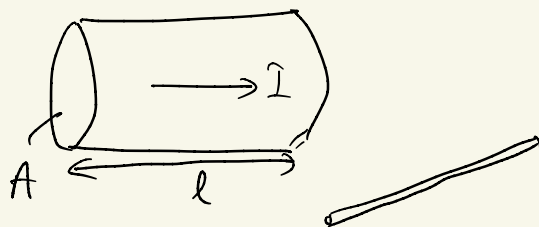
just for future, you'll see!

Resistance

best conductor: Ag (silver)

$$R = \rho \frac{l}{A}$$

resistivity
(material property)

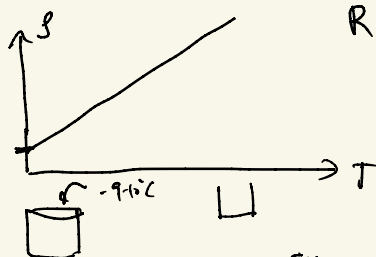


$$R = \rho \frac{l}{A}$$

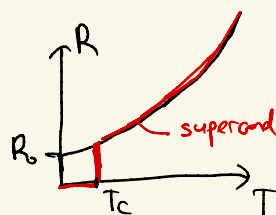
ρ vs T :

$$\rho = \rho_0 (1 + \alpha(T - T_0))$$

$$\rho - \rho_0 = \alpha(T - T_0)$$



Superconductors

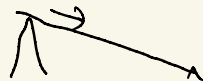
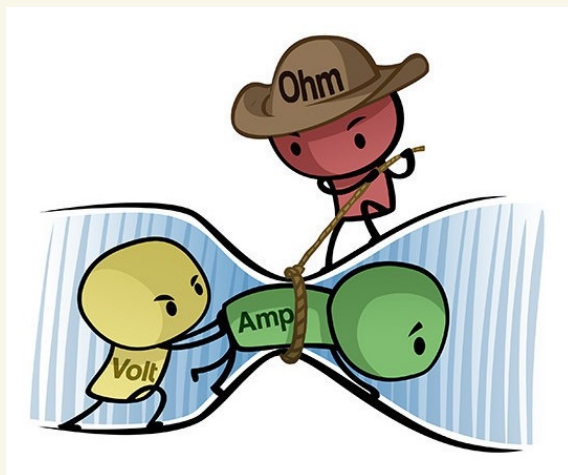


Nb, Al: 2K, 10K

Закон Ома

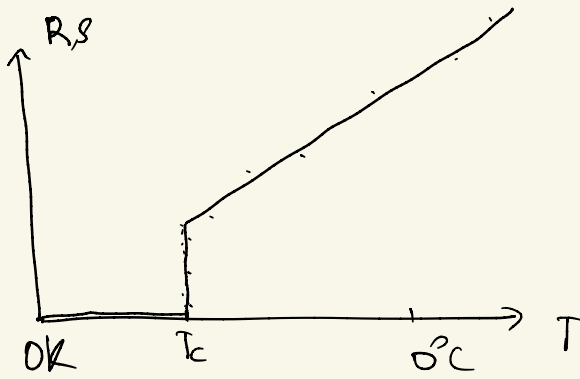
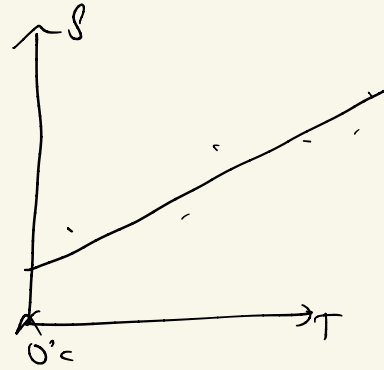
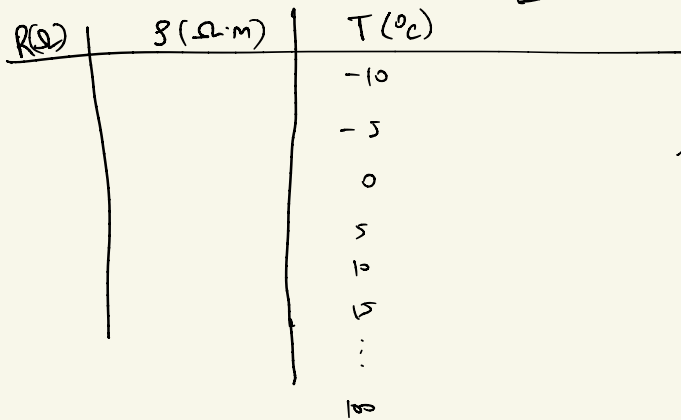
$$V = IR$$

Ohm's law: current is always directly proportional to ΔV potential diff. applied to the device (wire).



$l, A (\text{m}^2)$

$$R = \frac{\rho l}{A} \rightarrow \rho = \frac{RA}{l}$$



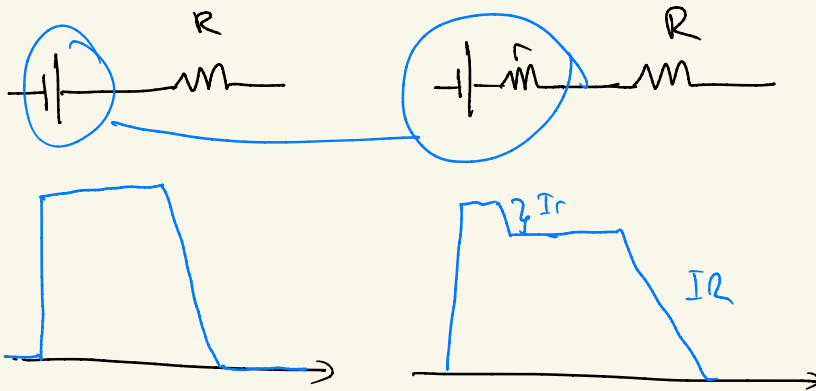
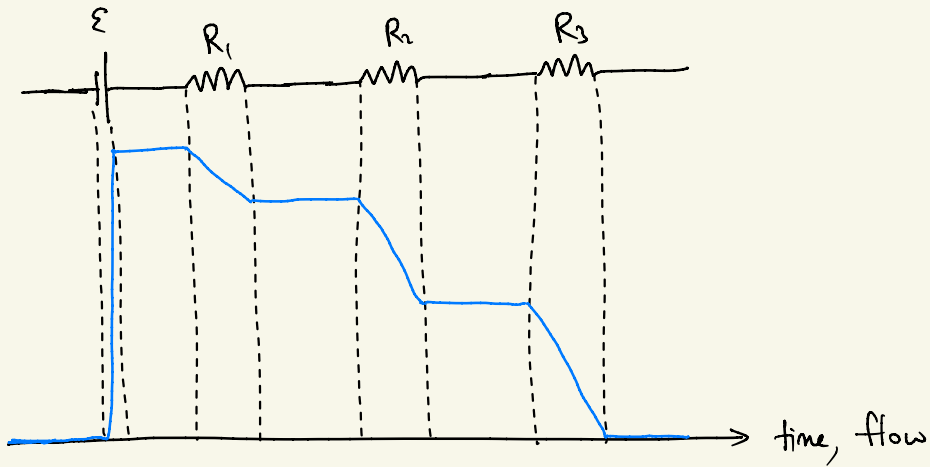
Al: 2K

Nb: 9.7K

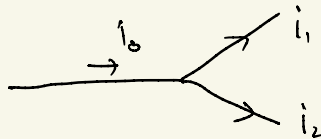
273K

liquid Nitrogen & dilution refrigerator (cryostat)

Chapter 27: Circuits (general)



Junction



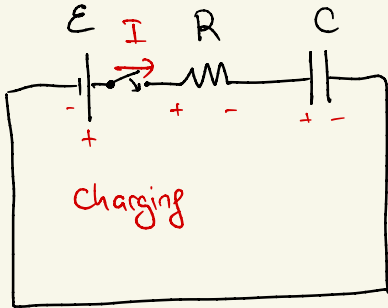
$$i_0 = i_1 + i_2$$

RC Circuits

- Used in high/low pass filters (electronics)
- Pulse shaping

Part 1: Charging

$$Q(t=0) = 0$$



$$V = \frac{q}{C}$$

$$\mathcal{E} - IR - \frac{q}{C} = 0$$

$$\mathcal{E} - \frac{dq}{dt}R - \frac{q}{C} = 0$$

$$\mathcal{E} - \frac{q}{C} = \frac{dq}{dt}R$$

$$\frac{dq}{(\mathcal{E} - \frac{q}{C})} = \frac{dt}{R}$$

$$\int_0^Q \frac{dq}{\mathcal{E}C - q} = \int_0^t \frac{dt}{RC}$$

Method of Substitution:

$$\mathcal{E}C - q = u$$

$$\frac{du}{dq} = -1 \rightarrow dq = -du$$

$$-\int \frac{du}{u} = \int_0^t \frac{dt}{RC}$$

$$\ln u \Big|_{u_i}^{u_f} = -\frac{t}{RC}$$

$$\ln[\mathcal{E}C - Q] - \ln(\mathcal{E}C) \quad \ln(\mathcal{E}C - q) \Big|_0^Q = -\frac{t}{RC}$$

$$\ln(\varepsilon C - Q) - \ln(\varepsilon C) = -\frac{t}{RC}$$

$$\ln A - \ln B = \ln\left(\frac{A}{B}\right)$$

$$e^{\ln x} = x$$

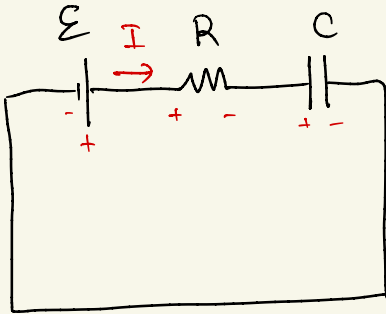
$$\ln\left(\frac{\varepsilon C - Q}{\varepsilon C}\right) = -\frac{t}{RC}$$

$$\frac{\varepsilon C - Q}{\varepsilon C} = e^{-\frac{t}{RC}}$$

$$\varepsilon C - Q = \varepsilon C e^{-\frac{t}{RC}}$$

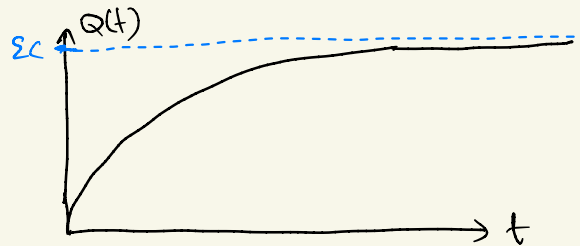
$$\varepsilon C - \varepsilon C e^{-\frac{t}{RC}} = Q(t)$$

$$Q(t) = \varepsilon C (1 - e^{-\frac{t}{RC}})$$



Let's observe $Q(t)$:

$$Q(t=0) = \varepsilon C (1 - e^0) = \varepsilon C (1 - 1) = 0$$



$$I = \frac{dQ(t)}{dt} = \frac{d}{dt}(\varepsilon C (1 - e^{-\frac{t}{RC}}))$$

$$\begin{aligned}
 I &= \frac{dQ(t)}{dt} = \frac{d}{dt}(\epsilon C (1 - e^{-t/RC})) \\
 &= \epsilon C \frac{d}{dt}(1 - e^{-t/RC}) \\
 &= \epsilon C (0 - (-\frac{1}{RC}) e^{-t/RC})
 \end{aligned}$$

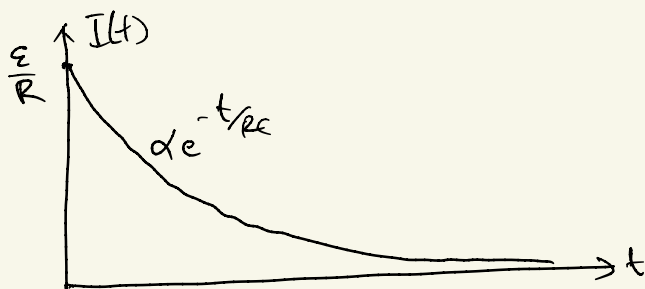
$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$= \frac{\epsilon C}{RC} e^{-t/RC}$$

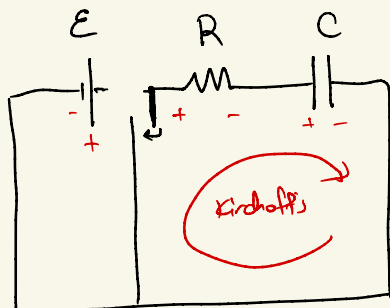
$$I(t) = \frac{\epsilon}{R} e^{-t/RC} \checkmark$$

$$V = IR$$

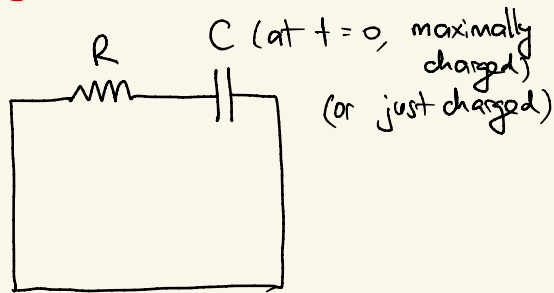
$$I = \frac{V}{R} \checkmark$$



Part 2: RC Circuits - discharging



$\Rightarrow$



$$-IR - \frac{q}{C} = 0$$

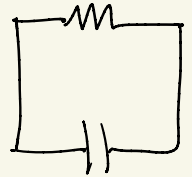
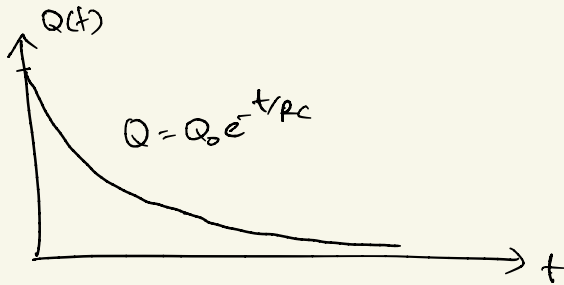
$$\frac{q}{C} + IR = 0$$

$$\frac{q}{C} + \frac{dq}{dt} R = 0 \rightarrow \int_Q^{q(t)} \frac{dq}{q} + \int \frac{RC}{dt} = 0$$

$$\ln q = -\frac{t}{RC}$$

$$\ln\left(\frac{Q(t)}{Q_0}\right) = -\frac{t}{RC}$$

$$Q(t) = Q_0 e^{-t/RC}$$

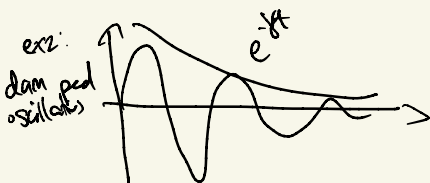


$$Q(t) = Q_0 e^{-t/RC} = Q_0 \left(e^{-t/\tau} \right) \quad \text{famous (well-known dependence)}$$

Time constant ($\tau = RC$) represents the time it takes for the voltage across the capacitor to reach

$$Q(t) \rightarrow \frac{Q_0}{e}$$

$$Q = CV$$



quality factor

$$N: A \rightarrow \frac{A}{e}$$

sec. 27-9 RC Circuits

•57 Switch S in Fig. 27-63 is closed at time $t = 0$, to begin charging an initially uncharged capacitor of capacitance $C = 15.0 \mu\text{F}$ through a resistor of resistance $R = 20.0 \Omega$. At what time is the potential across the capacitor equal to that across the resistor?

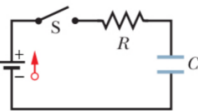


Fig. 27-63
Problems
57 and 96.

Let's write potentials across $\left\{ \begin{array}{l} \text{resistor: } V_R = I(t)R \\ \text{capacitor: } V_C = \frac{q(t)}{C} \end{array} \right.$

$t = ?$ $V_R = V_C$ $\longleftrightarrow$

We've already derived:

$$Q(t) = \varepsilon C (1 - e^{-t/RC})$$

&

$$I(t) = \frac{dQ(t)}{dt} = \frac{\varepsilon}{R} e^{-t/RC}$$

$$V_R(t) = V_C(t)$$

$$I(t)R = \frac{q(t)}{C}$$

$$\left(\frac{\varepsilon}{R} e^{-t/RC} \right) R = \frac{\varepsilon C (1 - e^{-t/RC})}{C}$$

$$e^{-t/RC} = 1 - e^{-t/RC}$$

$$2e^{-t/RC} = 1$$

$$\frac{1}{2} = e^{-t/RC}$$

Take log of both sides

$$\ln\left(\frac{1}{2}\right) = -\frac{t}{RC}$$

Also use

$$\ln A - \ln B = \ln\left(\frac{A}{B}\right)$$

$$\ln 1 - \ln 2 = -\frac{t}{RC}$$

$$e^x = 1 \rightarrow (e^0)$$

$$t = RC \ln 2$$

•65 In Fig. 27-66, $R_1 = 10.0 \text{ k}\Omega$, $R_2 = 15.0 \text{ k}\Omega$, $C = 0.400 \text{ }\mu\text{F}$, and the ideal battery has emf $\mathcal{E} = 20.0 \text{ V}$. First, the switch is closed a long time so that the steady state is reached. Then the switch is opened at time $t = 0$. What is the current in resistor 2 at $t = 4.00 \text{ ms}$?

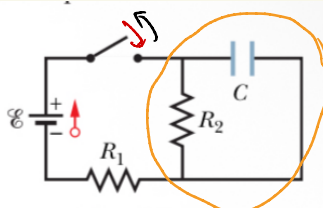
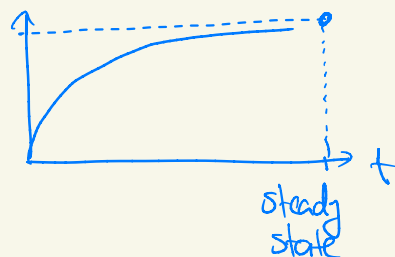


Fig. 27-66
Problems 65 and 99.

step 1:
Steady state: $q(t) = \mathcal{E}C(1 - e^{-t/RC})$
здесь - это когда конденсатор полностью заряжен.

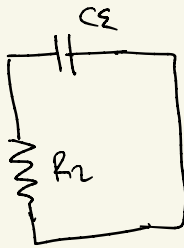


step 2: at $t = 0$, switch is opened
capacitor will start discharging.



из этого мы знаем что
discharging time const $\tau = R_2 C$

это нам даст rate of discharge, but we need to know initial charge/voltage across capacitor to get numerical value of $I(t)$.



$$q(t) = q_0 e^{-t/R_2 C}$$

$$q(t) = e^{-t/R_2 C}$$

=

=

$$e^{-\frac{4 \times 10^{-3}}{15000 \Omega \cdot 0.4 \mu F}}$$

at $t = 4 \text{ ms} = 4 \times 10^{-3} \text{ s}$

$$C = 0.4 \mu\text{F} = 0.4 \times 10^{-6} \text{ F}$$

$$R_2 = 15 \text{ k}\Omega$$

$$I(t) = \frac{dq}{dt} = I_0$$

$$e^{-t/R_2 C}$$

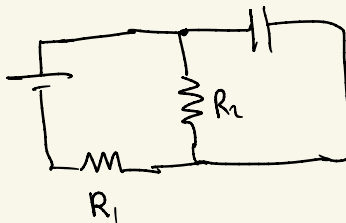
65. In the steady state situation, the capacitor voltage will equal the voltage across $R_2 = 15 \text{ k}\Omega$:

$$V_0 = R_2 \frac{\mathcal{E}}{R_1 + R_2} = (15.0 \text{ k}\Omega) \left(\frac{20.0 \text{ V}}{10.0 \text{ k}\Omega + 15.0 \text{ k}\Omega} \right) = 12.0 \text{ V}.$$

Now, multiplying Eq. 27-39 by the capacitance leads to $V = V_0 e^{-t/RC}$ describing the voltage across the capacitor (and across $R_2 = 15.0 \text{ k}\Omega$) after the switch is opened (at $t = 0$). Thus, with $t = 0.00400 \text{ s}$, we obtain

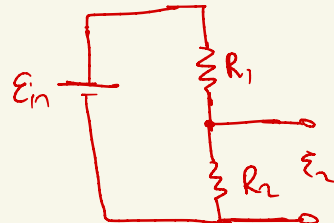
$$V = (12) e^{-0.004 / (15000)(0.4 \times 10^{-6})} = 6.16 \text{ V}.$$

Therefore, using Ohm's law, the current through R_2 is $6.16 / 15000 = 4.11 \times 10^{-4} \text{ A}$.



$$\frac{V_0}{R_2} = \frac{\mathcal{E}}{R_1 + R_2}$$

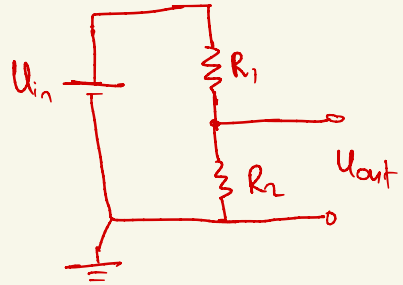
Voltage divider



$$\frac{V_o}{R_2} = \frac{\Sigma}{R_1 + R_2}$$

Voltage divider

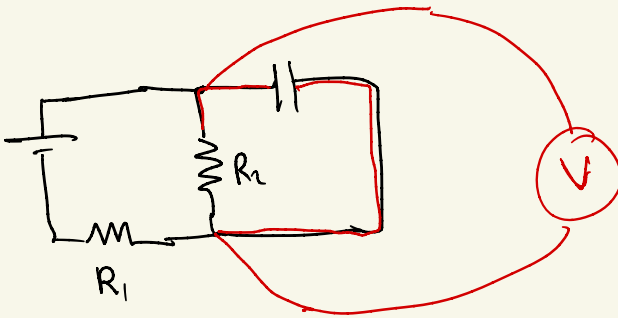
$$U_{out} = \frac{R_2}{R_1 + R_2} U_{in}$$



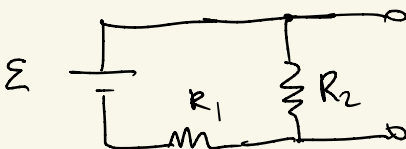
$$U_{in} = I(R_1 + R_2)$$

$$U_{out} = IR_2$$

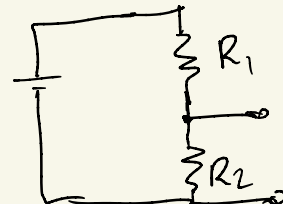
$$U_{out} = IR_2 = \left(\frac{U_{in}}{R_1 + R_2} \right) R_2 = \frac{R_2}{R_1 + R_2} U_{in}$$

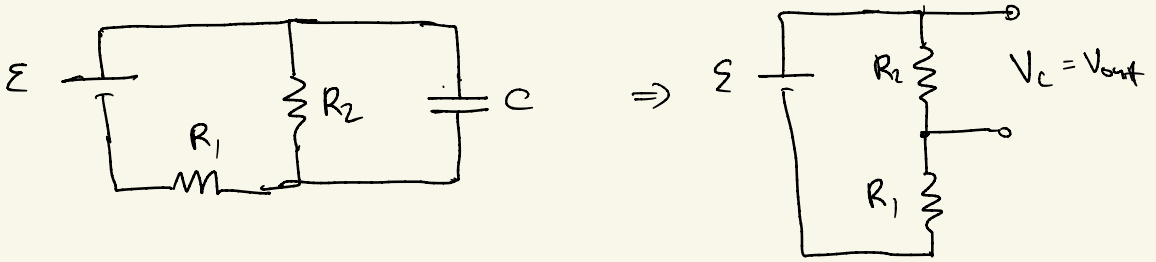


$$V_{R_2} = V_C$$



=



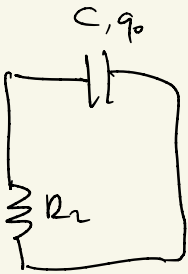


$$V_c = IR_2$$

$$\varepsilon = I(R_1 + R_2) \rightarrow I = \frac{\varepsilon}{R_1 + R_2}$$

$$V_c = IR_2 = \frac{\varepsilon}{R_1 + R_2} R_2$$

Discharging:



$$\begin{aligned} q(t) &= q_0 e^{-t/R_2 C} \\ &= V_c C e^{-t/R_2 C} \\ &= \frac{R_2 \varepsilon C}{R_1 + R_2} e^{-t/R_2 C} \end{aligned}$$

$$I(t) = \frac{dq(t)}{dt} = \left(\frac{R_2 \varepsilon C}{R_1 + R_2} \right) \frac{d}{dt} (e^{-t/R_2 C})$$

$$= - \frac{R_2 \varepsilon C}{(R_1 + R_2) R_2 C} e^{-t/R_2 C}$$

minus sign here appeared
because I is decreasing

so $I(t) = -\frac{dq}{dt}$ technically

$$= - \left(\frac{\varepsilon}{R_1 + R_2} \right) e^{-t/R_2 C}$$

HW Ch 26

Problems: 10, 11, 35, 36, 75

HW Ch 27

Problems: 4, 21, 23, 30, 31, 35, 36, 42, 48,
50, 57, 58, 59, 63, 65, 66, 80, 87

Deadline: Aug 17, Sunday 12:00 PM noon